Bijaya Hatuwal

Ph.D. Candidate in Computer Science (Expected May 2027)
Department of Electrical Engineering and Computer Science, University of Missouri, Columbia, MO, USA
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Profile

Final-year Ph.D. candidate in Computer Science researching data-efficient, knowledge-guided, and robust AI for 3D, volumetric, multi-channel, and multimodal data. Experienced Graduate Teaching Assistant for undergraduate and graduate computer science courses.

Research Interests: Deep Learning · Computer Vision · Medical Image Segmentation · 3D/Volumetric and Aerial Image Analysis · Self/Weak-Supervised Learning · Knowledge-Guided Learning · Multimodal Learning and Foundation Models · Generative AI · Computational Disease Modeling · Hybrid Quantum-Classical Machine Learning

Education

Ph.D. in Computer Science Expected May 2027
University of Missouri, Columbia, MO, USA
Advisor: Dr. Kannappan Palaniappan, Curators' Distinguished Professor, CIVA Lab
Dissertation: *Self-Supervised Learning for 3D Volume Segmentation*

M.S. in Computer Science Dec. 2024
University of Missouri, Columbia, MO, USA

B.E. in Computer Engineering 2018
Himalaya College of Engineering, Tribhuvan University, Nepal

Teaching Experience

Graduate Teaching Assistant Aug. 2024 – Present
Department of EECS, University of Missouri, Columbia, MO, USA
Support instruction for undergraduate and graduate courses through office hours, assignment design, and grading.

- CMP_SC 4450/7450: Principles of Programming Languages (undergraduate/graduate) Fall 2024, 2025, 2026
- CMP_SC 3330: Object-Oriented Programming Spring 2026
- CMP_SC 4320: Software Engineering I Spring 2025

Research Experience

Graduate Research Assistant Aug. 2021 – Present
University of Missouri, Columbia, MO, USA

- Design and evaluate deep learning methods for disease detection, segmentation, object tracking, and volumetric analysis of biomedical and aerial images (CIVA Lab, Department of EECS).
- Analyzed genomic data on disease transmission and evolution with biologists, including Bayesian phylogeographic analysis of Influenza A virus sequences.

Publications

Peer-Reviewed Conference/Workshop Papers

[1] K. Safavigerdini, **B. Hatuwal**, A. Daghighi, and K. Palaniappan, "Generative AI video evaluation: Survey of metrics, benchmarks, and trustworthiness," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5189–5200, 2026.

[2] C. Algire, A. Kalukin, H. Theiss, **B. Hatuwal**, T. Krock, S. Israel, and K. Palaniappan, "Shape fidelity: Measuring a digital model's trueness to form," in *Proc. SPIE*, vol. 14032, p. 140320Q, 2026.

[3] **B. Hatuwal**, R. Bao, M.-L. Ho, and K. Palaniappan, "HE3D-Net: Hypoxic Ischemic Encephalopathy diagnosis and lesion segmentation using 3D heterogenous ensemble models," in *Applied Imagery Pattern Recognition Workshop (AIPR)*, Springer, pp. 495–507, 2025.

[4] M. Kristan, J. Matas, P. Tokmakov, M. Felsberg, L. Č. Zajc, A. Lukežič, et al. (incl. **B. Hatuwal**), "The second visual object tracking segmentation VOTS2024 challenge results," in *European Conference on Computer Vision (ECCV) Workshops*, Springer, pp. 357–383, 2024.

[5] E. Soltanikazemi, A. Dhakal, **B. Hatuwal**, I. E. Toubal, A. Aboah, and K. Palaniappan, "Real-time helmet violation detection in AI City Challenge 2023 with genetic algorithm-enhanced YOLOv5," in *IEEE Applied Imagery Pattern Recognition Workshop (AIPR)*, 2023.

Journal Articles

[6] V. Goel, J. Ding, **B. Hatuwal**, E. Giri, T. J. DeLiberto, J. Lowe, R. Webby, M. Emch, and X.-F. Wan, "Ecological drivers of evolution of swine Influenza in the United States: A review," *Emerging Microbes & Infections*, vol. 14, no. 1, p. 2455598, 2025.

[7] R. K. Raghavan, F. B. Osei, A. Stein, S. Hesting, L. Jaster, **B. Hatuwal**, J. E. Mosley, and A. Raghavan, "Evaluation of spatial variation in chronic wasting disease risk with Bayesian Poisson log-Gaussian model," *Frontiers in Veterinary Science*, vol. 12, p. 1568468, 2025.

[8] **B. Hatuwal**, V. Goel, T. J. DeLiberto, J. Lowe, M. Emch, R. J. Webby, and X.-F. Wan, "Spatial patterns of Influenza A virus spread across compartments in commercial swine farms in the United States," *Emerging Microbes & Infections*, vol. 13, no. 1, p. 2400530, 2024.

[9] C. Gao, F. Wen, M. Guan, **B. Hatuwal**, L. Li, B. Praena, C. Y. Tang, J. Zhang, F. Luo, H. Xie, et al., "MAIVeSS: Streamlined selection of antigenically matched, high-yield viruses for seasonal Influenza vaccine production," *Nature Communications*, vol. 15, no. 1, p. 1128, 2024.

[10] **B. Hatuwal**, A. Shakya, and B. Joshi, "Plant leaf disease recognition using random forest, KNN, SVM and CNN," *Polibits*, vol. 62, pp. 13–19, 2020.

[11] **B. Hatuwal** and H. C. Thapa, "Lung cancer detection using convolutional neural network on histopathological images," *International Journal of Computer Trends and Technology*, vol. 68, no. 10, pp. 21–24, 2020.

Presentations

- Oral Presentation, "HE3D-Net: Hypoxic Ischemic Encephalopathy Diagnosis and Lesion Segmentation Using 3D Heterogenous Ensemble Models," Applied Imagery Pattern Recognition Workshop (AIPR), Medical and Biomedical Imaging Track, 2025.
- Poster Presentation and Flash Talk, American Society for Virology Workshop, Columbus, OH, June 2024.
- Oral and Poster Presentation, 19th Mexican International Conference on Artificial Intelligence (MICAI), Mexico City, Mexico (virtual), October 2020.

Service, Governance, and Leadership

President 2026 – Present
Nepalese Student Association (MUNSA), University of Missouri

Secretary 2025 – 2026
Nepalese Student Association (MUNSA), University of Missouri

Vice President Aug. 2024 – July 2025
Graduate Student Association of Electrical Engineering and Computer Science (EECS-GSA), University of Missouri

Fellowships, Training, and Certifications

- EECS Departmental Travel Fellowship, University of Missouri, October 2025.
- Road to Practitioner Program, IBM, August 2026.
- Quantum Computing Training (Qiskit), IBM: quantum algorithms, quantum-classical workflows, error mitigation, variational algorithms, and quantum machine learning.
- FAA Part 107 Commercial Remote Pilot Certificate (sUAS); sUAS Commercial Remote Pilot Training, Kansas State University Salina Aerospace and Technology Campus.

Industry Experience: Information Systems and Data Engineering

Software Engineer May 2020 – July 2021
Cotiviti, Kathmandu, Nepal

- Built and maintained data pipelines, transformation workflows, and reporting systems for U.S. health data across providers and insurance (MS-SQL/T-SQL, SSIS, SSRS, Tableau).

Data Engineer Nov. 2019 – Mar. 2020
Impetus Incorporation Pvt. Ltd., Kathmandu, Nepal

- Developed data integration, transformation, and analytics workflows.

Associate Software Engineer (Intern, Jan. to Mar. 2019) Jan. 2019 – Oct. 2019
LIS Nepal Pvt. Ltd., Lalitpur, Nepal

- Developed ETL and retail data-warehousing solutions (SQL Server, BIRT, Oracle BI Publisher) for clients including Makro China, Express (Bamboo Rose, USA), and the Nepal Stock Exchange (NEPSE).

Technical Skills

Programming	Python, Java, C/C++, SQL
Machine Learning	PyTorch, MONAI, scikit-learn, Vision Transformers, Swin Transformer, Ensembles, Natural Language Processing (NLP), Self/Weak-Supervised Learning
Data & Systems	ETL/ELT pipelines, MS-SQL/T-SQL, SSIS, SSRS, Tableau; GPU/HPC cluster computing

Professional Memberships

- American Society for Virology (ASV), 2024 to 2025
- Mexican Society of Artificial Intelligence (SMIA), 2020 to 2021

References

Available upon request.

Last updated: November 2026